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Original Research Article

PFAS uptake by lettuce: Roles of PFAS properties, co-present organic acids, and nanoplastics

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ABSTRACT

The accumulation of per- and polyfluoroalkyl substances (PFAS) in food crops poses a significant food safety and human health risk, yet studies on the accumulation of PFAS in food crops in conditions that more realistically represent a growth environment with various co-occurring chemicals are still lacking. This study investigated the accumulation of perfluorooctanoic acid (PFOA), perfluorooctane sulfonic acid (PFOS), and perfluorobutane sulfonic acid (PFBS) by lettuce in a hydroponic system in the co-presence of polystyrene nanoplastics (NPs, 300 nm) and/or three prevalent organic acids [humic acid (HA), fulvic acid (FA), and oxalic acid (OA)]. Results showed that NPs decreased the concentration of all three PFAS compounds in lettuce shoots after seven days of exposure compared with plants exposed to PFAS alone. However, the three organic acids all increased PFAS concentrations in lettuce shoots, regardless of the presence of NPs. Three interesting trends were observed: (1) long-chain PFAS (PFOA, PFOS) appeared to depend more heavily on the active transport pathway than short ones; (2) the properties of organic acids play a major role in their impact on PFAS accumulation in lettuce; and (3) organic acids showed greater impact on the accumulation of short-chain PFAS (PFBS) in lettuce shoots than the long ones. Overall, the co-presence of organic acids increased PFAS accumulation in lettuce, suggesting that interactions with organic acids and nanoplastics can materially alter PFAS uptake. These findings highlight the need for additional mechanistic and soil-based studies to better understand PFAS food-safety risks under realistic field conditions.

1. Introduction

Per- and polyfluoroalkyl substances (PFAS) have become a major societal concern due to their widespread presence, extreme stability, and strong immunotoxicity and carcinogenicity [1]. Soil is a well-known sink for a wide variety of pollutants, and agricultural soils are particularly prone to the accumulation of pollutants due to the routine introduction of various agrichemicals. While PFAS has never been intentionally introduced to agricultural lands, decades of biosolid application, irrigation with PFAS-contaminated water, and extensive use of PFAS-containing pesticides have markedly raised PFAS levels in agricultural soils in the USA, according to several recent studies [2–5]. For example, repeated biosolid application has been reported to elevate PFAS concentrations in agricultural soils to approximately 65−170 μg/kg, which is 100 times higher than the global background level of PFAS [2]. The elevated concentrations of PFAS in agricultural soil are extremely concerning due to their propensity for plant uptake, which provides a significant pathway for PFAS to enter the human food chain and threaten human health.

Previous studies have shown that PFAS display distinctive bioaccumulation patterns in plants as compared with other legacy organic contaminants. For neutral organic compounds, the octanol-water partition coefficient (log K_{ow}) is a dominant factor governing their plant uptake; however, log K_{ow} does not provide a reliable indication for the PFAS plant uptake due to the ionization of some PFAS in circumneutral pH [6,7]. In general, PFAS exhibits relatively high root-to-shoot transfer and bioaccumulation in plant shoots [8]. In addition to the passive transport, PFAS can cross root cell membranes and eventually reach the xylem tissue via various protein transporters and ion channels, giving PFAS a higher potential for plant accumulation. Plasma membrane H^+-ATPase, anion channels, and aquaporins have been demonstrated to play important roles in PFAS plant root uptake [9–11].

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Importantly, plant PFAS uptake is also influenced by surrounding environmental factors, including the presence of co-contaminants and nutrient management strategies [12,13]. Among the various factors that may alter plant PFAS uptake are the increasing levels of micro- and nanoplastics (MNPs), defined as plastic particles with a size smaller than 5 mm and 1 μm, respectively [14]. MNPs can be unintentionally introduced into the soil together with PFAS via biosolid application and irrigation [15]. Elevated MNP concentrations in biosolids have been reported in multiple previous studies. For instance, up to 287,000 particles/kg of dry biosolid sludge were reported in an earlier study [16]. It is important to note that several commonly used techniques for MNP detection in soil lack the ability to detect nanoscale plastic particles, suggesting that biosolids likely contain much higher numbers of plastic particles than previously reported. When biosolids are land applied, MNPs are introduced to agricultural soils. In addition, with the widespread adoption of plastic mulch and films in agricultural practices, a substantial amount of MNPs in soil are generated from their weathering and remain in soil for a significant period of time [17]. The persistence of MNPs and their co-existence with PFAS in agricultural soils has been increasingly recognized [16,18]. These MNPs vary in composition, shape, and size. Some can accumulate in plant tissues themselves and pose risks to human health [19]; however, the greater risk of MNPs in agricultural soil may arise from their interactions with co-existing environmental pollutants, leading to elevated plant uptake and accumulation of these co-contaminants. Previous studies have shown that co-present MNPs increased the accumulation of cadmium (Cd^{2+}) [20] and neutral organic contaminants [21,22]. However, how MNPs may alter the uptake of PFAS by plants has rarely been studied. This is critical because most commonly detected PFAS, including perfluoroalkyl carboxylic acids (PFCAs) and perfluoroalkyl sulfonic acids (PFSAs), are typically present as anions in the plant-growth environment. Therefore, the interactions between these anionic PFAS with MNPs are expected to be very different from those between MNPs and cationic metal ions and neutral organic compounds. Additionally, PFAS contains a wide variety of compounds with different chain lengths, molecular structures, and head functional groups. How these properties affect the interaction of PFAS and MNPs and eventually their plant accumulation is still unknown.

Natural organic matter (NOM) is ubiquitous in plant-growth environments. Even though NOM is rarely considered in the plant uptake of environmental pollutants, a few previous studies have shown that some NOMs, such as humic acid (HA), may modify the plant physiological responses, including the expression of plant protein transporters [23]. More importantly, emerging evidence suggests that NOM can remarkably alter PFAS accumulation in plants [24,25]. Liu et al. showed that HA facilitated the cross-membrane transport of several PFAS compounds across wheat root cells by activating the expression of slow-type anion channels [24]. Separately, Guo et al. demonstrated that fulvic acid (FA) enhanced the accumulation of 6:2 chlorinated polyfluoroalkyl ether sulfonate (6:2 Cl-PFESA) in plants by stimulating the activity of plasma membrane H^+-ATPase in wheat root cells while HA inhibited it [26]. This contrasts with the observation by Canellas et al. who showed that HA enhanced the activity of H^+-ATPase in maize root [23]. Despite these discrepancies, existing literature clearly suggests that NOM can significantly alter plant PFAS uptake, and such an impact should not be ignored. Previous studies also suggested that the NOM impact on plant PFAS accumulation likely varies with NOM properties, plant species, and PFAS compounds. In addition to widespread NOMs, plant roots excrete significant amounts of root exudates containing a large number of low-molecular-weight organic acids. A previous study showed that oxalic acid (OA) in the root exudate of lettuce greatly enhanced the accumulation of perfluorooctanoic acid (PFOA) in plants [27]. The authors attributed this increase to the reduced adsorption of PFOA onto soil particles, which increased the bioavailability of PFOA. However, whether OA would directly affect plant physiology and cause altered PFAS plant uptake like other organic acids was not investigated. Considering the high possibility of the co-occurrence of PFAS, MNP and NOMs and the various potential interactions among them that may markedly alter the plant uptake and human exposure to PFAS, it is imperative to investigate how the co-existence of organic acids and MNPs modifies plant PFAS accumulation.

The objectives of our study were to (1) examine PFAS property-dependent plant uptake; (2) determine the impact of co-present NPs and organic acids on the plant accumulation of PFAS in a binary system, and (3) evaluate the combined effect of NPs and organic acids on plant PFAS uptake in a ternary system. All studies were performed in hydroponic systems to minimize the compounding effect of soils, allowing the impact of co-present organic acids and NPs on plant uptake of PFAS to be directly observed. Lettuce was chosen as a model plant because it is a common leafy vegetable in the human diet and is one of the most commonly used plant species in plant PFAS uptake studies [28,29]. Polystyrene nanoplastics (PSNPs) were used as representative NPs because they are commonly detected in agricultural soil and are frequently used in plant NP studies [30].

2. Materials and methods

2.1. Chemicals and plant materials

Sodium carbonate (Na_2CO_3) was purchased from Alfa Aesar (MA, USA). Suwannee River fulvic and humic acids were purchased from Ambeed (IL, USA). OPTIMA LCMS-grade ammonium hydroxide (NH_4OH) and methanol (MeOH); ACS-grade glacial acetic acid; and sodium hydroxide (NaOH) were purchased from Fisher Chemical (NH, USA). Tetrabutylammonium hydrogen sulfate (TBAHS) was purchased from MP Biomedicals (OH, USA). PSNPs with a diameter of 300 nm, PFOA, perfluorooctane sulfonic acid (PFOS), and perfluorobutane sulfonic acid (PFBS) were purchased from Sigma Aldrich (MO, USA). Denatured ethanol (EtOH) containing 5% methanol and tert-butyl methyl ether (MTBE) were purchased from Thermo Scientific (TX, USA). Modified Hoagland salts were purchased from US Biological (MA, USA). ACS-grade hydrochloric acid (HCl), oxalic acid, phosphate-buffered saline (PBS), polysorbate (Tween) 20, sodium phosphate dibasic (Na_2HPO_4), and Whatman GF-D membranes, 0.2 μm nitrocellulose membranes were purchased from VWR (PA, USA). Oasis HLB 6 cc (200 mg) SPE cartridges were purchased from Waters (MA, USA).

Lactuca sativa (Fusion M1 Lettuce) seeds were obtained from Johnny's Selected Seeds (ME, USA). They were sown on moistened filter paper for two days under dark conditions and two days under simulated long day conditions (16:8 h light:dark cycle, 24–30 °C, 40%–70% relative humidity) until cotyledons were developed and radicles were at least 1 cm long. Seedlings were transplanted to foil-wrapped Falcon tubes containing 1/4 Hoagland solution and grown for 28 days under germination conditions.

2.2. Uptake of PFAS by lettuce in the presence of organic acids and nanoplastics

Roots of 28-day-old lettuce seedlings were washed with ultrapure water, then transferred to new Falcon tubes with different exposure solutions. A total of nine treatments were prepared: Ultrapure water (CK), solution containing 100 μg/L of each of PFOA, PFOS and PFBS (PFAS), PFAS treatment with 10 mg/L humic acid (PFAS + HA), PFAS treatment with 10 mg/L fulvic acid (PFAS + FA), and PFAS treatment with 1 mg/L oxalic acid (PFAS + OA). For treatments containing PFAS, a parallel set of treatments containing 50 mg/L of NPs were also prepared, creating a new binary or ternary system that includes PFAS, organic acids and nanoplastics. Those treatments were labeled as PFAS + NPs, PFAS + NPs + HA, PFAS + NPs + FA, and PFAS + NPs + OA, respectively. Plants were exposed to the treatment solutions for seven days before harvest. Water lost through transpiration was replenished every other day with deionized water and was recorded. Three replicates were

prepared for each treatment. In the setup, PFAS and PSNP concentrations were chosen based on several previous studies showing no overt toxicity of these chemicals at the chosen levels and their analytical detection limits of in plant tissues. Concentrations of HA and FA were chosen for their environmental relevance, which typically fall within the range of 2–15 mg/L in natural water. OA concentration was chosen based on its concentration in typical root exudates [31].

2.3. Measurements of plant physiological parameters

After seven days of exposure, the youngest leaf with a circular area of at least 1 cm^2 was analyzed for chlorophyll fluorescence using an OS1p fluorometer (Opti-Sciences, NH, USA). Afterward, the seedlings were removed from the growing media and washed with excess ultrapure water and air-dried. Root and shoot tissues were then separated at the crown of the plant with a sharp pair of scissors. Fresh weights of each tissue were recorded. About 100 mg of fresh leaf tissue was collected for chlorophyll analysis following well-established methods [32]. Briefly, chlorophyll was extracted in 4 mL dimethyl formamide (DMF) at 4 °C overnight, and the absorbance of the solution was measured at 664 and 647 nm using a Lambda 35 UV-vis spectrophotometer (PerkinElmer, MA, USA). The root and remaining leaf tissues were dried in an oven at 60 °C until constant weight, roughly 24 h, and then the dry weight was recorded.

2.4. PFAS analysis in plant tissues

PFAS analysis in lettuce tissues followed a previously reported method [33]. Briefly, the dried tissues were crushed with a metal spatula in 50 mL Falcon tubes, and 4 mL of 0.4 M NaOH was added to each sample tube. The samples were then digested overnight at 4 °C. The following day, 2 mL of 0.5 M TBAHS and 4 mL of 0.25 M Na_2CO_3 were added to the sample tubes and vortexed to homogenize the digested tissue and solution. Afterward, 5 mL of MTBE was added to the sample tubes, and the tubes were shaken at 300 rpm for 20 min. The MTBE and aqueous layers were separated by centrifugation at 4500×*g* for 4 min. The MTBE layer was then transferred to a new 15 mL centrifuge tube. This extraction process was repeated twice more. MTBE was evaporated from the 15 mL centrifuge tubes under a constant N_2 stream at 60 °C using a ReactiVap (Thermo Fisher, TX, USA). The samples were then reconstituted using 10 mL of 10% MeOH. Final solid phase extraction (SPE) was performed on the sample using Oasis HLB SPE cartridges preconditioned with 10 mL of MeOH, followed by 10 mL of MTBE. After the reconstituted samples passed through the cartridge, PFAS was eluted using 4 mL of MeOH and 4 mL of 0.1% NH_4OH in MeOH. Solutions were again evaporated as before, then reconstituted in 1 mL MeOH. Reconstituted samples were transferred to 1.8 mL HPLC vials and analyzed by liquid chromatography-tandem mass spectrometry (LC-MS/MS) (Thermo Fisher Scientific, Waltham, USA) following EPA method 1633. For quality control/quality assurance (QC/QA), a mass-labeled internal standard ($^{13}C_8$-PFOA) was injected into the extracted samples before LC-MS/MS analysis. In addition, a blank was analyzed every 8 samples to ensure that the column was not contaminated by residues from previous samples. The limit of quantification (LOQ), determined based on a signal-to-noise ratio of 5:1, was 0.25 ng/mL. PFAS concentrations in plant tissue were reported as ng PFAS/g dry tissue weight. The translocation factor (TF) was calculated as the concentration of PFAS in the shoot divided by the concentration in the root.

2.5. Inhibitor study

To elucidate the transporters involved in active PFAS cross-membrane transport, solutions of various inhibitors were prepared according to Table 1. Roots of 28-day-old lettuce plants were washed with ultrapure water, then transferred to new Falcon tubes containing the inhibitor solutions overnight. The plant seedlings were then thoroughly rinsed with deionized water and exposed to the treatment solution containing 100 μg/L of each of the three PFAS compounds (PFOA, PFOS and PFBS) for 24 h. Three replicates were prepared for each inhibitor solution, and PFAS in plant roots was analyzed as described above after harvest.

Table 1
Summary of the chemical inhibitors and their target transporters in this study.

Name	Code	Concentration (μM)	Transporter affected
Control	CK	-	-
Sodium Orthovanadate	SV	500	H^+-ATPase [34]
Silver Nitrate	AG	1	Aquaporins [35]
Anthracene-9-carboxylic acid	9AC	10	Slow-type anion channels [36]
Diisothiocyanostilbene-2,2-Disulfonic acid	DIDS	3.125	Rapid-type anion channels [31]
Tetrabromobenzotriazole	TBB	10	Ca^{2+}-dependent protein kinases [37]

2.6. Statistics

All data were reported as mean ± standard deviation. Comparisons among multiple treatments were conducted using one-way ANOVA followed by a post hoc Tukey's test, while comparisons between two treatments were performed using a two-tailed Student's *t*-test. Statistical significance was set at $\alpha = 0.05$. Different lowercase letters indicate significant differences among groups ($p < 0.05$).

3. Results and discussion

At the tested concentration, none of the PFAS treatments caused conspicuous signs of toxicity to lettuce plants. The accumulative water transpiration (Fig. S1) and the dry biomass of lettuce tissues (Fig. S2) from different treatments both indicated minimal obvious toxicity of PFAS at the treatment level. However, the exposure to PFAS treatment alone significantly lowered the total chlorophyll content compared to the control (Fig. S3). The presence of HA mitigated the effect of PFAS, increasing the total chlorophyll content of PFAS-treated lettuce to a similar level as the control plant, while the other two organic acids had insignificant effects. Surprisingly, the PFAS + NPs + HA and PFAS + NPs + OA treatments significantly lowered the total chlorophyll content of lettuce compared with the PFAS-alone treatment. However, these physiological differences did not translate into differences in photosynthetic efficiency among these plants (Fig. S4). Our results suggested that PFAS toxicity is unlikely to be a major factor for the varied PFAS accumulation in lettuce from different treatments, as detailed below, even though the exposure to these different combinations of PFAS, NPs, and organic acids can cause subtle physiological responses at the molecular level, which require further, more detailed studies.

3.1. PFAS plant uptake: Role of chain length and head functional group

Trace levels of PFOA and PFOS were detected in several control plants; however, PFBS was never detected in the control plants. These trace levels of long-chain PFAS were likely from cross-contamination of shared lab facilities and exposure to air-borne PFAS because several other studies on PFAS were ongoing during this study. However, the PFAS in control plants was extremely low and did not affect the interpretation of results in PFAS-treated plants. After exposure to the PFAS treatment containing 100 μg/L of each of the three PFAS (PFOA, PFOS, and PFBS) for seven days, they were all detected in both plant root and shoot tissues (Fig. 1).

PFOS accumulation in lettuce root was the lowest compared to the other two PFAS compounds but remained within a comparable range

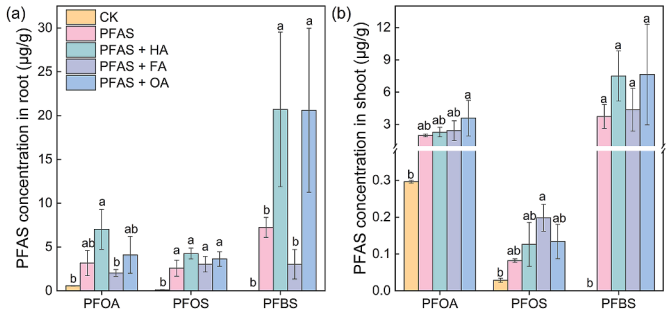


Fig. 1. PFAS concentrations in lettuce roots (A) and shoots (B) in the presence of different organic acids. Data are means ± SD ($n = 3$). Different lowercase letters above bars indicate significant differences among treatments for a specific PFAS compound. CK, control; HA, humic acid; FA, fulvic acid; OA, oxalic acid.

(Fig. 1a). For instance, the PFOS concentration in lettuce roots was approximately 81.8% of the PFOA concentration and 45.8% of the PFBS concentration. However, its concentration in lettuce shoot was orders of magnitude lower than the other two PFAS compounds (Fig. 1b), indicating minimal *in-planta* transfer. Interestingly, PFBS showed not only the highest root and shoot concentration among PFAS compounds, but also the highest root to shoot transfer, as indicated by the TF (0.66) (Table S1). The result agrees with the general consensus that short-chain PFAS are more prone to plant accumulation in aboveground tissues than long-chain PFAS, and PFAS containing a carboxylic acid head functional group tend to have higher accumulation than PFAS with the sulfonic acid functional group. Previous studies suggested that both passive cross-membrane transport with the transpiration stream and active transport via various ion channels and protein transporters such as H^+-ATPase and slow-type anion channels contribute to PFAS plant root uptake and *in-planta* transfer [26,35]. However, most previous studies focused on a few high-profile long-chain PFAS compounds, and information is still scarce concerning the transporters involved in short-chain PFAS, hindering the mechanistic understanding of the plant uptake of PFAS with different chain lengths.

3.2. Impact of co-present organic acids on plant PFAS uptake

Plant uptake of PFAS is not solely governed by the inherent properties of PFAS. Numerous previous studies have reported that environmental factors, including the presence of NOM, can considerably alter the plant uptake of various environmental chemicals [24]. Our results are consistent with several previous studies that demonstrated a notable impact of NOM on the uptake of PFOA and several other PFAS by wheat [24,26]. Our results further demonstrated that the impact of co-existing organic acids on plant uptake of PFAS varied with both the properties of the organic acids and the properties of PFAS (Fig. 1a). The co-presence of HA increased the root concentration of all three PFAS compounds. The most pronounced increase was observed for PFBS (+260%), followed by PFOA (+122%) and PFOS (+64.2%). In contrast, FA only slightly increased the root concentration of PFOS (+17.2%) but decreased the root concentration of PFOA (−35.7%) and PFBS (−46.4%). These substantial and remarkably different effects of HA and FA on the plant root uptake of PFAS showed that the co-present natural organic acids and their characteristics should be carefully considered in assessing plant accumulation of PFAS in the natural environment. To our best knowledge, this is the first study to demonstrate the pronounced impact of natural organic acids on short-chain PFAS (e.g., PFBS). Our results also showed for the first time that OA, a low-molecular-weight organic acid and a common component of plant root exudates, increased the root concentration of all three PFAS compounds, even though the increase was smaller than that observed for HA (Fig. 1a). The biggest enhancive effect for OA was observed for PFBS (+265%) followed by PFOS (+40.3%) and PFOA (+29.6%). The varying impact of organic acids on the root uptake of different PFAS suggests potentially different pathways for these PFAS to enter lettuce root cells.

To gain more insights into the cross-membrane transport of different PFAS, a transporter inhibitor experiment was conducted. In general, both DIDS and TBB showed negligible impact on the root accumulation of two long-chain PFAS, but insignificantly increased the concentration of PFBS by 69.9% and 58.2%, respectively (Fig. 2), suggesting that the rapid-type anion channels and Ca^{2+}-dependent protein kinases are not involved in PFAS cross-membrane transport of long-chain PFAS, but might play an important role in short-chain PFAS. In comparison, SV, the H^+-ATPase inhibitor, markedly lowered the root concentration of all three PFAS, with reductions of 62.7% for PFOA, 73.1% for PFOS, and 65.1% for PFBS, suggesting a substantial role of H^+-ATPase in plant uptake of both short-chain and long-chain PFAS. AG, an aquaporin inhibitor, also decreased the root concentration of PFOA and PFOS by 32.7% and 51.0%, but insignificantly increased PFBS concentration by 36.2%, implying that aquaporins may regulate the long-chain and short-chain PFAS differently. The 9AC inhibitor study further demonstrated the striking differences between long-chain and short-chain PFAS with regard to their root accumulation mechanisms. 9AC, a slow-type anion channel inhibitor, decreased the PFOA and PFOS root concentrations by 16.3% and 20.8%, respectively, but increased the PFBS root concentration by 10.3%. Overall, the inhibitor study suggests a greater dependence of active transport for long-chain PFAS than short-chain PFAS, and that plants regulate their entrance differently. We postulate that this phenomenon may reflect the differences in hydrophobicity of long-chain and short-chain PFAS, with the greater hydrophobicity of long-chain PFAS making them more likely to bind to the membrane lipid bilayer and less likely to pass through root cell membranes passively than their short-chain counterparts. The hydrophobicity-dependent cross-membrane transport of organic contaminants has been well recognized in the literature. However, it is important to caution that the inhibitor study only provides indirect evidence on the involvement of specific transporters in PFAS cross-membrane transport. Additional investigation involving a broader range of PFAS and direct molecular analysis of gene and protein expressions of potential protein and ion transporters are needed to clarify the different root absorption pathways of PFAS with varying chain lengths.

The three organic acids also showed a distinctive impact on the PFAS concentration in lettuce shoots (Fig. 1b). In general, HA showed greater impact on short-chain PFAS than on long-chain PFAS, and a stronger

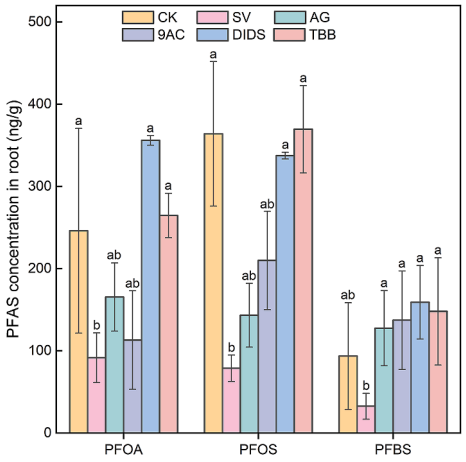


Fig. 2. PFAS concentrations in lettuce roots after plant seedlings were exposed to different inhibitors. Data are means ± SD ($n = 3$). Different lowercase letters above bars indicate significant differences among treatments for a specific PFAS compound. SV, sodium orthovanadate; AG, silver nitrate; 9AC, anthracene-9-carboxylic acid; DIDS, Diisothiocyanostilbene-2,2-Disulfonic acid; TBB, Tetrabromobenzotriazole.

effect on PFSAs than on PFCAs. For instance, HA increased the PFBS concentrations in lettuce shoots by 100%, while it only increased the PFOS and PFOA concentrations by 53.4% and 15.5%, respectively. Similar to the relatively milder impact of FA on PFAS root concentration, FA also displayed a relatively insignificant impact on PFAS concentrations in lettuce shoots (from +16.6% for PFBS to +22.1% for PFOA). However, FA markedly increased the PFOS concentration in lettuce shoots by 142% compared to plants exposed to PFAS alone. The co-presence of OA increased the concentration of all three PFAS in lettuce shoots, showing in general a greater impact on short-chain PFAS (+104% for PFBS) than long-chain PFAS (+80.9% for PFOA and +63.2% for PFOS). Because OA is a component of plant root exudates that vary considerably between different plant species, it is unclear whether what was observed for lettuce in this study applies to other plant species that excrete different low-molecular-weight organic acids. For lettuce, however, the elevated PFAS concentrations in the presence of different organic acids are concerning because, with the increasing use of short-chain PFAS, our results suggest that they may pose a greater risk to food safety than long-chain PFAS due to their greater tendency to accumulate in plant tissues. Future research should expand to other low-molecular-weight acids common in plant root exudates, which may shed light on the potential mechanisms for varying PFAS accumulation potential of different plant species.

A further examination of the PFAS concentration in both the root and shoot tissues and the TF indicated that the underlying mechanisms by which co-present organic acids affect PFAS plant accumulation varied with the organic acids and PFAS (Table S1). For instance, the elevation of PFAS concentration in lettuce shoots by HA was attributed to an even greater increase of PFAS concentration in lettuce roots because HA actually decreased the TF. This might be due to the formation of the HA-PFAS complex, which is more readily adsorbed on the root surface but more difficult to enter root cells and reach the vascular tissue [36]. In contrast, FA markedly improved the *in-planta* root-to-shoot transfer of all three PFAS, leading to higher accumulation in lettuce shoots even though it decreased some PFAS concentrations in lettuce roots. The varying impacts of the three organic acids imply diverse uptake mechanisms of different PFAS compounds, in agreement with the potentially varying cross-membrane transfer mechanisms for long-chain and short-chain PFAS as revealed by the inhibitor study.

3.3. Impact of nanoplastics on plant PFAS uptake

In the system containing both PFAS and NPs, the co-presence of NPs lowered the concentration of all three PFAS compounds in both lettuce root and shoot tissues (Fig. 3), although the decrease was insignificant for the two long-chain PFAS in lettuce roots. The TF of PFAS in this binary system indicated that NPs did not negatively affect the root-to-shoot transport (Table S1). Instead, the reduced PFAS accumulation was primarily attributed to the decreased PFAS accumulation in plant roots. The precise mechanism for the lower PFAS concentration in lettuce roots caused by NPs is yet to be determined, but we postulate that the adsorption of PFAS on NPs lowered the bioavailability of PFAS in our study due to the relatively large size of NPs (300 nm), much larger than the root cell membrane pores of uncompromised fresh roots. This is supported by a previous study showing that most NPs were attached to the root surface of Arabidopsis and wheat and could be removed by the washing procedures prior to the root analysis [38]. It is also possible that NPs have modified the expression of protein transporters related to PFAS absorption because several previous studies have shown that NPs can modify the expression of some plant proteins associated with oxidative stress and root regulation [39], suggesting that exposure to NPs can lead to molecular-level changes in plants. The modification of aquaporin genes by polyvinyl chloride (PVC) MNPs has been reported recently [18]. However, the exact effect of polystyrene NPs on transporter proteins involved in PFAS root absorption is still poorly understood. Direct evidence on the NP-induced changes in the expression of PFAS transporters will shed more light on the roles that NPs play in plant PFAS uptake. Once PFAS entered the vascular tissues, their transport appeared unhindered by NPs, as evidenced by the comparable TF of PFAS compounds in the presence and absence of NPs (Table S1). It should be mentioned that the NPs used in this study were around 300 nm, which is smaller than the diameter of most vascular channels in plants. How NPs of different sizes and shapes may affect PFAS plant uptake differently requires additional studies.

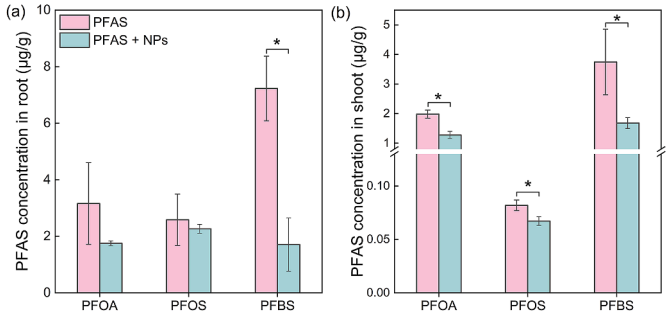


Fig. 3. PFAS concentrations in lettuce roots (A) and shoots (B) in the presence of polystyrene nanoplastics (NPs). Data are means ± SD ($n = 3$). Star above bars indicates significant difference based on Student's t-test ($p < 0.05$).

3.4. PFAS plant uptake in the co-presence of nanoplastics and organic acids

The actual growing environment of plants often contains a variety of constituents; however, their collective impact on plant uptake of PFAS has never been examined. Compared with plant PFAS uptake in the presence of NPs alone, the co-presence of organic acids invariably enhanced PFAS accumulation in both lettuce roots and shoots in the PFAS-NPs-organic acid ternary system (Fig. 4). In general, all three organic acids displayed greater impact on the accumulation of short-chain PFBS in lettuce tissues than long-chain PFAS. For instance, HA increased the concentration of PFBS by 698% while it only increased the concentration of PFOA and PFOS by 174% and 85.2% in lettuce roots. Similar trends were observed for PFAS in lettuce shoots, where all three organic acids increased PFAS concentration, with a greater effect observed for PFBS. Intriguingly, while HA and FA have shown notable differences in their impact on PFAS accumulation in the PFAS + organic acid binary system, these two acids appeared to have remarkably similar impacts on the accumulation of PFAS in lettuce shoots in the ternary system. For example, in the ternary system, both HA and FA increased the PFBS concentration in lettuce shoots by about 420% and increased the other two long-chain PFAS concentrations by about 102% to 145%. The impact of OA was consistently smaller than the other two acids in the ternary system, increasing PFBS concentration in lettuce shoots by

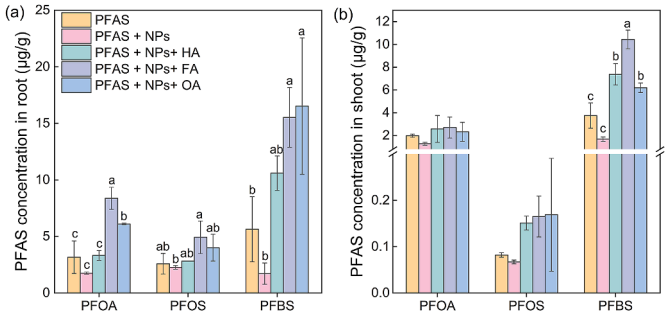


Fig. 4. PFAS concentrations in lettuce roots (A) and shoots (B) in the presence of both polystyrene nanoplastics (NPs) and organic acids. Data are means ± SD ($n = 3$). Different lowercase letters above bars indicate significant differences among treatments for a specific PFAS compound.

269%, PFOS by 151%, and PFOA by 81.9%. Clearly, the presence of NPs has modified the interactions of organic acids and PFAS in the system. The results suggest that complex interactions between PFAS, NPs, and organic acids markedly affect the final accumulation of PFAS in plant tissues. We postulate that the formation of NP-organic acid complexes might play a notable role in the more comparable impact of HA and FA in the ternary system than in the binary system. The distinctively different behavior of OA from the other two organic acids in the ternary system reflects its distinctive physicochemical properties, such as its much smaller size. Additional studies are critically needed to reveal the underlying processes governing the interactions of PFAS, NPs, and co-existing natural organic acids and the final fate of PFAS in a plant system.

Importantly, even though plant PFAS uptake has been investigated for years, most studies were carried out in ideal laboratory settings in which no interfering chemicals were present. Our results confirmed previous reports that co-present organic acids and NPs can appreciably alter plant PFAS uptake. It is particularly noteworthy that the co-presence of NPs and organic acids invariably increased the PFAS concentration in both lettuce roots and shoots in the ternary system to a level that was higher than in plants exposed to PFAS alone on average. While studies on PFAS uptake without interferences are critical for the mechanistic understanding of PFAS plant interactions, results obtained from ideal settings may underestimate the real risk of PFAS to food safety and human health. The ternary system investigated in this study represents a realistic scenario of plant cultivation due to the ubiquity of organic acids and growing accumulation of NPs in their growing environment. Our results underscored the importance of assessing the PFAS plant accumulation potential in a more realistic environment, and caution must be exercised to extrapolate greenhouse results obtained in ideal growing conditions to field settings.

CRediT authorship contribution statement

Michael Taylor Bryant: Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Data curation. **Lorenzo Rossi:** Writing – review & editing. **Nihong Wen:** Writing – review & editing. **Xingmao Ma:** Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

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